

## a Study design

### Benchmarks 160 tasks



#### BixBench-Verified-50

50 questions · Answer  
Evaluator acceptance, 0 or 1



#### CompBioBench

100 questions · Answer  
Reconstructed-key agreement, 0 or 1



#### IWC

10 workflows · Output files  
Workflow-output agreement, 0–1

### Experimental design

Runs in

#### Custom code

Isolated workspace

Tools

Software the agent installs

Prompt

Task

Time limit

120 min†; 6 or 12 h‡

#### Galaxy

usegalaxy.org via MCP

Galaxy tools and UDTs\*

Task + Galaxy policy

6 or 12 h‡

\*Not on IWC †CompBioBench ‡IWC; none stated otherwise

#### Four model configurations (Codex agent)



GPT-5.5



GPT-5.6 Sol



GPT-5.6 Luna



DeepSeek V4 Pro



3 independent runs per condition



$$160 \times 4 \times 2 \times 3 = 3,840$$

tasks model configs conditions runs primary runs

3,840 runs scored on 160 tasks; 4,240 archived runs include configurations outside this comparison

### Evaluation



#### Accuracy

Benchmark-specific score; not pooled



#### Repeatability

Same outcome in all 3 runs?



#### Input tokens

Per run, including cached



#### Galaxy interface

Requests · failures · UDTs



#### Failure causes

AI-assisted audit of failures